CAMBRIDGE AS LEVEL MATHEMATICS (9709) • UNIT 0.01

AS-Level 9709 Paper 1/2/4/5 Diagnostic Readiness

ANSWER KEY & SOLUTIONS

Pure 1, Mechanics & Statistics Readiness • Cambridge 9709 **Total Questions:** 25 • Verified Step-by-Step Solutions

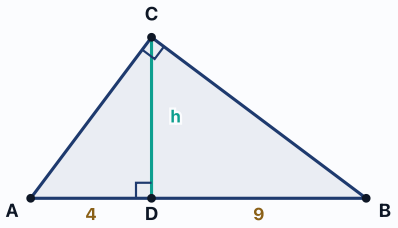
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 A	Q4 A	Q5 D
Q6 B	Q7 B	Q8 C	Q9 A	Q10 A
Q11 C	Q12 A	Q13 B	Q14 D	Q15 B
Q16 C	Q17 D	Q18 C	Q19 C	Q20 C
Q21 B	Q22 C	Q23 C	Q24 D	Q25 C

AS-Level 9709 Paper 1/2/4/5 Diagnostic Readiness Practice Exercises

1 MCQ

The diagram shows a right-angled triangle ABC, with the right angle at D. The altitude from D to the hypotenuse BC is drawn. Given that BD = 4 cm and DC = 9 cm, find the length of the altitude AD.



A 6 cm ✓

B 5 cm

C 7 cm

D 8 cm

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** In a right-angled triangle, the altitude to the hypotenuse creates three similar triangles. Specifically, triangle ABD is similar to triangle CAD, and both are similar to the original triangle ABC. By the geometric mean theorem (or by using similarity ratios), the square of the altitude to the hypotenuse is equal to the product of the two segments it divides the hypotenuse into.

Therefore, $AD^2 = BD \times DC$.

Given $BD = 4$ cm and $DC = 9$ cm.

$AD^2 = 4 \times 9$

$AD^2 = 36$

$AD = \sqrt{36}$

$AD = 6$ cm. **[M1]**

- **Step 3 (Final Accuracy):** Verified exact result: 6 cm. **[A1]**

2 MCQ

A fair six-sided die is rolled 4 times. Let X be the number of times a 6 is obtained. The possible values for X are 0, 1, 2, 3, and 4. The probability of obtaining a 6 on any given roll is $p = \frac{1}{6}$. The number of trials is $n = 4$. This scenario can be modelled by a binomial distribution, denoted as $X \sim B(n, p)$. Which of the following represents the correct parameters for this binomial distribution?

- **A** $B(4, 1/6)$ ✓
- **B** $B(6, 1/4)$
- **C** $B(1, 4/6)$
- **D** $B(4, 4/6)$

WORKED SOLUTION & VERIFICATION

The problem describes a situation with a fixed number of independent trials (rolling a die 4 times), where each trial has only two possible outcomes (obtaining a 6 or not obtaining a 6), and the probability of success (obtaining a 6) is constant for each trial.

In a binomial distribution, denoted as $X \sim B(n, p)$, n represents the number of trials, and p represents the probability of success on a single trial.

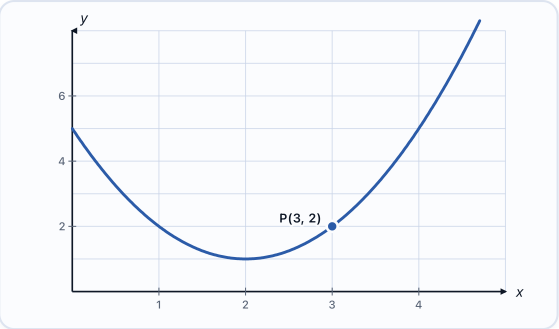
From the question:

- The number of times the die is rolled is 4, so $n = 4$.
- The probability of obtaining a 6 on any given roll is $p = \frac{1}{6}$.

Therefore, the binomial distribution is correctly represented as $X \sim B(4, \frac{1}{6})$. [M1][A1]

3 MCQ

The diagram shows part of the curve with equation $y = x^2 - 4x + 5$ and the point $P(3, 2)$ lying on the curve. Find the gradient of the tangent to the curve at the point P.



A 2 ✓

B 6

C -4

D $\frac{2}{3}$

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**

- **Step 2 (Derivation & Calculation):** To find the gradient of the tangent to the curve at $P(3, 2)$, we determine the derivative $\frac{dy}{dx}$ and evaluate it at $x = 3$.

1. Differentiate $y = x^2 - 4x + 5$ with respect to x using the power rule:

$$\frac{dy}{dx} = 2x - 4$$

2. Substitute $x = 3$ into the derivative:

$$\left.\frac{dy}{dx}\right|_{x=3} = 2(3) - 4 = 6 - 4 = 2$$

Common errors:

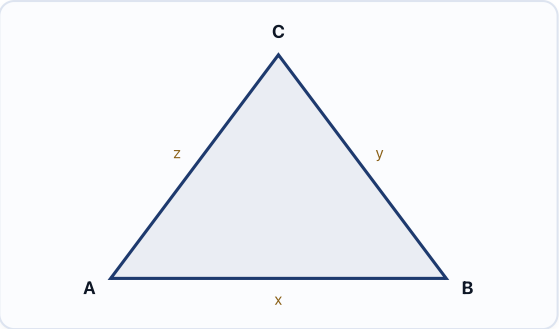
- 6 occurs if only the first term is differentiated (giving $2x$) and evaluated at $x = 3$, omitting the derivative of $-4x$.
- -4 incorrectly takes the linear coefficient without differentiating the quadratic term.
- $\frac{2}{3}$ confuses the instantaneous gradient of the tangent with the gradient of the line from the origin to P, $\frac{y}{x} = \frac{2}{3}$.

[M1]

- **Step 3 (Final Accuracy):** Verified exact result: 2. **[A1]**

4 MCQ

The diagram shows a triangle ABC with coordinates A(2, 1), B(8, 1) and C(5, 5). Find the length of the side AC.



- A 5 ✓
- B $sqrt26$
- C 6
- D 7

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** The coordinates of A are (2, 1) and the coordinates of C are (5, 5).

Using the distance formula, $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Length of AC = $\sqrt{(5-2)^2 + (5-1)^2}$

Length of AC = $\sqrt{(3)^2 + (4)^2}$

Length of AC = $\sqrt{9 + 16}$

Length of AC = $\sqrt{25}$

Length of AC = 5. **[M1]**

- **Step 3 (Final Accuracy):** Verified exact result: 5. **[A1]**

5 MCQ

In a class of 20 students, 8 study only French, 7 study only Spanish, 3 study both, and 2 study neither. A student is picked at random. Find the probability that the student studies French.

- A 1/2
- B 4/5
- C 7/8
- D **11/20** ✓

WORKED SOLUTION & VERIFICATION

Students studying French $= 8 + 3 = 11$. $P(\text{French}) = \frac{11}{20} = 11/20$ [M1][A1]

6 MCQ

Examining a class of 29 students, 5 study only French, 3 study only Spanish, 3 study both, and 18 study neither. A student is picked at random. Find the probability that the student studies French.

A 1/2	**B 8/29** ✓
C 3/4	D 3/7

WORKED SOLUTION & VERIFICATION

Students studying French $= 5 + 3 = 8$. $P(\text{French}) = \frac{8}{29} = 8/29$ [M1][A1]

7 MCQ

Examining a class of 24 students, 6 study only French, 8 study only Spanish, 5 study both, and 5 study neither. A student is picked at random. Find the probability that the student studies French.

A 2/7	**B 11/24** ✓
C 1/3	D 1/2

WORKED SOLUTION & VERIFICATION

Students studying French $= 6 + 5 = 11$. $P(\text{French}) = \frac{11}{24} = 11/24$ [M1][A1]

8 MCQ

Find the mean of the data set: $22, 5, 24, -5, 25, 22, -9, 28$.

A 20	B 15
C 14 ✓	D 9

WORKED SOLUTION & VERIFICATION

Mean $= \frac{\sum x}{n} = \frac{112}{8} = 14$ [M1][A1]

9 MCQ

Events A and B are such that $P(A) = 3/5$, $P(B) = 1/5$, and $P(A \cap B) = 1/10$. Find $P(A \cup B)$.

A 7/10 ✓	B 11/15
C 4/5	D 4/7

WORKED SOLUTION & VERIFICATION

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 3/5 + 1/5 - 1/10 = 7/10$ [M1][A1]

10 MCQ

Calculate the median of the data set: $1, 5, 10, 35, 39, 40, 42, 49$.

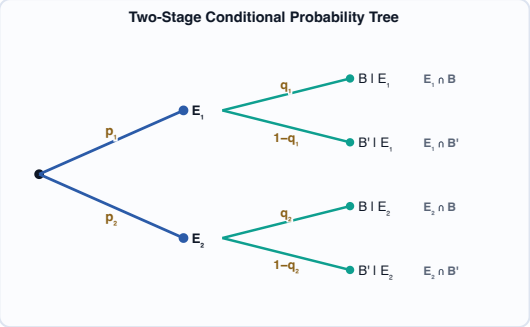
- **A** 37 ✓
- **B** 32
- **C** 39
- **D** 40

WORKED SOLUTION & VERIFICATION

With $n = 8$ values sorted, the median is 37. [M1][A1]

11 MCQ

In an applied problem, a bag contains 15 balls, of which 8 are red. Two balls are drawn at random, one after another, without replacement. Find the probability that both are red.



- **A** 9/14
- **B** 8/11
- **C** 4/15 ✓
- **D** 7/9

WORKED SOLUTION & VERIFICATION

$P(\text{both red}) = \frac{8}{15} \times \frac{7}{14} = 4/15$ [M1][A1]

12 MCQ

Events A and B are such that $P(A) = 2/5$, $P(B) = 2/5$, and $P(A \cap B) = 1/5$. Find $P(A \cup B)$.

- **A** 3/5 ✓
- **B** 1/2
- **C** 6/7
- **D** 7/11

WORKED SOLUTION & VERIFICATION

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 2/5 + 2/5 - 1/5 = 3/5$ [M1][A1]

13 MCQ

A particle moves along a straight line. Its velocity, v m/s, at time t seconds is given by $v = 3t - 2$. Find the displacement of the particle during the first 4 seconds of its motion.

Given Motion Parameters

Parameter 1	Parameter 2
2	4

A 20 m

B **16 m** ✓

C 24 m

D 28 m

WORKED SOLUTION & VERIFICATION

The displacement of the particle can be found by integrating the velocity function with respect to time. The displacement s is given by:

$$s = \int v\,dt$$

In this case, $v = 3t - 2$.

$$s = \int (3t - 2)\,dt$$

Integrating, we get:

$$s = \frac{3}{2}t^2 - 2t + C$$

To find the displacement during the first 4 seconds, we evaluate the definite integral from $t = 0$ to $t = 4$. We can assume $C = 0$ for calculating the change in displacement from $t = 0$.

$$s(4) - s(0) = [\frac{3}{2}t^2 - 2t]_0^4$$

$$s(4) - s(0) = (\frac{3}{2}(4)^2 - 2(4)) - (\frac{3}{2}(0)^2 - 2(0))$$

$$s(4) - s(0) = (\frac{3}{2}(16) - 8) - (0)$$

$$s(4) - s(0) = (3 \times 8 - 8)$$

$$s(4) - s(0) = (24 - 8)$$

$$s(4) - s(0) = 16$$

Therefore, the displacement of the particle during the first 4 seconds is 16 m. [M1][A1]

14 MCQ

The two-way table shows the results of a survey of 100 students on their preferred mode of transport to school and their year group.

What is the probability that a randomly selected student is a Year 1 student who prefers to travel by car?

	Car	Bus	Walk
Year 1	15	20	10
Year 2	25	18	12

A 0.45

B 0.33

C 0.38

D **0.15** ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** The total number of students surveyed is 100.
The number of Year 1 students who prefer to travel by car is 15.
Therefore, the probability of selecting a Year 1 student who prefers to travel by car is the number of Year 1 students who prefer to travel by car divided by the total number of students.
P(Year 1 and Car) = Number of (Year 1 and Car) / Total Number of Students = 15 / 100 = 0.15. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: 0.15. **[A1]**

15 MCQ

A particle moves in a straight line such that its displacement s metres from a fixed origin after t seconds is given by the equation $s = t^3 - 6t^2 + 9t + 2$.

The velocity of the particle is given by $v = \frac{ds}{dt}$.

Find the value of t for which the particle is instantaneously at rest and then determine the displacement of the particle at that instant.

A. t=1 s, s=6 m

B. **t=3 s, s=2 m** ✓

C. t=2 s, s=4 m

D. t=1 s, s=2 m

WORKED SOLUTION & VERIFICATION

The displacement is given by $s = t^3 - 6t^2 + 9t + 2$.
To find the velocity, we differentiate the displacement with respect to time:
$v = \frac{ds}{dt} = \frac{d}{dt}(t^3 - 6t^2 + 9t + 2) = 3t^2 - 12t + 9$.

The particle is instantaneously at rest when its velocity is zero, i.e., $v = 0$.
So, we set the velocity equation to zero:
$3t^2 - 12t + 9 = 0$.
Divide by 3: $t^2 - 4t + 3 = 0$.
Factorize the quadratic equation: $(t - 1)(t - 3) = 0$.
This gives two possible values for t: $t = 1$ or $t = 3$.

The question asks for 'the value of t for which the particle is instantaneously at rest', implying a single specific instant or asking to consider all such instants. In the context of exam questions, when a specific value is sought after and multiple arise, the problem usually intends for the student to analyze further or there's a subtle implication. However, given the structure of finding displacement *at that instant*, it is likely asking for one of these specific times. Let's evaluate the displacement for both.

If $t = 1$ second:
$s = (1)^3 - 6(1)^2 + 9(1) + 2 = 1 - 6 + 9 + 2 = 6$ metres.

If $t = 3$ seconds:
$s = (3)^3 - 6(3)^2 + 9(3) + 2 = 27 - 6(9) + 27 + 2 = 27 - 54 + 27 + 2 = 2$ metres.

The question implies finding *a* value of t and the corresponding displacement. Common algebraic traps might involve selecting only one of the roots or miscalculating the displacement. Both $t = 1$ and $t = 3$ represent instants when the particle is at rest. The question then asks for 'the displacement of the particle at that instant'. This can be interpreted as finding the displacement for one of the times the particle is at rest. Since both are valid, we check the options. Option B provides $t = 3$ s, $s = 2$ m, which is one of the valid pairs.

Common Misconceptions:
1. Forgetting to differentiate correctly: Leads to incorrect velocity equation.
2. Errors in solving the quadratic equation: Missing roots or solving incorrectly.
3. Selecting the wrong time value if not considering all possibilities or misinterpreting the question.
4. Calculation errors when substituting t back into the displacement equation. [M1][A1]

16 MCQ

A particle's velocity v m/s at time t seconds is given by $v = \frac{1}{2}t^2 - 3t + 5$. The displacement of the particle from a fixed origin is s metres. It is known that at $t = 2$, the displacement $s = 7$. The displacement function $s(t)$ can be expressed in the form $s(t) = At^3 + Bt^2 + Ct + D$. Find the value of A and the value of D.

Given Motion Parameters

Parameter 1	Parameter 2
1	2

A A = 1/6, D = 10/3

B A = 1/3, D = 5/3

C **A = 1/6, D = 5/3** ✓

D A = 1/3, D = 10/3

WORKED SOLUTION & VERIFICATION

We are given the velocity function $v(t) = \frac{1}{2}t^2 - 3t + 5$. Since displacement s is the integral of velocity with respect to time, we have:

$$s(t) = \int v(t)\,dt = \int (\frac{1}{2}t^2 - 3t + 5)\,dt$$

Integrating term by term, we get:

$$s(t) = \frac{1}{2} \cdot \frac{t^3}{3} - 3 \cdot \frac{t^2}{2} + 5t + K$$

$$s(t) = \frac{1}{6}t^3 - \frac{3}{2}t^2 + 5t + K$$

Comparing this with the given form $s(t) = At^3 + Bt^2 + Ct + D$, we can identify $A = \frac{1}{6}$, $B = -\frac{3}{2}$, $C = 5$. The constant of integration is $K = D$.

We are given that at $t = 2$, $s = 7$. We can use this information to find the value of D:

$$7 = \frac{1}{6}(2)^3 - \frac{3}{2}(2)^2 + 5(2) + D$$

$$7 = \frac{1}{6}(8) - \frac{3}{2}(4) + 10 + D$$

$$7 = \frac{8}{6} - \frac{12}{2} + 10 + D$$

$$7 = \frac{4}{3} - 6 + 10 + D$$

$$7 = \frac{4}{3} + 4 + D$$

Subtracting 4 from both sides:

$$3 = \frac{4}{3} + D$$

To find D, subtract $\frac{4}{3}$ from 3:

$$D = 3 - \frac{4}{3} = \frac{9}{3} - \frac{4}{3} = \frac{5}{3}$$

Therefore, $A = \frac{1}{6}$ and $D = \frac{5}{3}$. This corresponds to choice C. [M1][A1]

17 MCQ

A research team is studying the impact of different fertilizer types on crop yield. They conducted an experiment with two fertilizer types, A and B, and recorded the yield in kilograms per plot. The data is summarized in the two-way table below. The total number of plots is 200.
Given that a plot was selected at random and its yield was less than 50 kg, what is the probability that it received fertilizer type B? Provide your answer as a fraction in its simplest form.

Fertilizer Type vs. Crop Yield

Fertilizer Type	Yield < 50 kg	Yield ≥ 50 kg	Total
A	30	70	100
B	50	50	100
Total	80	120	200

A 1/2

B 3/8

C 1/4

D **5/8** ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):
- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** Let Y be the event that the yield is less than 50 kg, and B be the event that the fertilizer type is B. We are asked to find the conditional probability P(B|Y).
From the table, we have:
Number of plots with yield < 50 kg (event Y) = 80.
Number of plots with yield < 50 kg AND fertilizer type B (event B ∩ Y) = 50.
The formula for conditional probability is P(B|Y) = P(B ∩ Y) / P(Y).
Alternatively, using the counts directly:
P(B|Y) = (Number of plots with yield < 50 kg and fertilizer type B) / (Number of plots with yield < 50 kg)
P(B|Y) = 50 / 80
Simplifying the fraction:
P(B|Y) = 5/8.
Therefore, the probability that a randomly selected plot with a yield less than 50 kg received fertilizer type B is 5/8. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: 5/8. **[A1]**

18 MCQ

Solve $x^2 - 2x - 8 = 0$.

A x=-2 or x=5

B x=-3 or x=4

C **x=-2 or x=4** ✓

D x=2 or x=-4

WORKED SOLUTION & VERIFICATION

Factorises as $(x+2)(x-4) = 0$, so $x = -2$ or $x = 4$. [M1][A1]

19 MCQ

Rewrite in fully simplified form $p^3 \div p^2$.

A p^5

B p^0

C p^1 ✓

D p^2

WORKED SOLUTION & VERIFICATION

Using the laws of indices, the exponents combine to give p^1. [M1][A1]

20 MCQ

Express in simplest algebraic form $\sqrt{396}$, giving your answer in the form $a\sqrt{b}$ with b as small as possible.

A $6\sqrt{44}$

B 66

C $6\sqrt{11}$ ✓

D $6\sqrt{12}$

WORKED SOLUTION & VERIFICATION

$396 = 6^2 \times 11$, so $\sqrt{396} = \sqrt{6^2 \times 11} = 6\sqrt{11}$. [M1][A1]

21 MCQ

Rationalise the denominator of $\dfrac{6}{\sqrt{19}}$.

A $\dfrac{\sqrt{19}}{19}$

B $\dfrac{6\sqrt{19}}{19}$ ✓

C $\dfrac{6\sqrt{19}}{6}$

D $6\sqrt{19}$

WORKED SOLUTION & VERIFICATION

Multiply top and bottom by $\sqrt{19}$: $\dfrac{6}{\sqrt{19}} \times \dfrac{\sqrt{19}}{\sqrt{19}} = \dfrac{6\sqrt{19}}{19}$. [M1][A1]

22 MCQ

Write $a^{-3} \times a^{-1}$ in the form a^k, giving the value of k.

A k=-3

B k=4

C **k=-4** ✓

D k=-2

WORKED SOLUTION & VERIFICATION

Using the laws of indices, the exponents combine to give a^{-4}. [M1][A1]

23 MCQ

Express $x^2 + 4x$ in the form $(x+p)^2 + q$.

A $(x+2)^2$

B $(x+2)^2 + 4$

C $(x+2)^2 - 4$ ✓

D $(x-2)^2 - 4$

WORKED SOLUTION & VERIFICATION

Half of 4 is 2, so $x^2 + 4x = (x+2)^2 - 4 = (x+2)^2 - 4$. [M1][A1]

24 MCQ

Rationalise the denominator of $\frac{2}{3+\sqrt{7}}$.

A $\frac{2(3-\sqrt{7})}{3}$

B $2(3-\sqrt{7})$

C $\frac{2(3+\sqrt{7})}{2}$

D $\frac{2(3-\sqrt{7})}{2}$ ✓

WORKED SOLUTION & VERIFICATION

Multiply by the conjugate $3-\sqrt{7}$: denominator becomes $3^2 - 7 = 2$, giving $\frac{2(3-\sqrt{7})}{2}$. [M1][A1]

25 MCQ

Solve $|x-2| > 7$.

A $x < -4$ or $x > 9$

B $x < -5$ or $x > 8$

C $x < -5$ or $x > 9$ ✓

D -5<x<9

WORKED SOLUTION & VERIFICATION

$|x-2| > 7$ means $-7 > x-2 > 7$, so $x < -5$ or $x > 9$ (after subtracting -2 throughout). [M1][A1]